

Getting started with Amazon Simple Storage Service (Amazon S3)

AWS Prescriptive Guidance

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance. This guide walks you through the fundamental concepts, storage classes, access control mechanisms, data management features, performance best practices, and cost-optimization strategies for Amazon S3. By the end, you will have the knowledge to architect robust, cost-efficient, and highly available data storage solutions on AWS.

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Introduction

Amazon S3 is one of the foundational services of the AWS Cloud, providing a simple web services interface to store and retrieve any amount of data from anywhere on the internet. Since its launch in 2006, Amazon S3 has grown to store trillions of objects and regularly peaks at millions of requests per second. It is used across virtually every industry for backup and restore, archiving, big data analytics, data lakes, media hosting, and static website hosting.

Amazon S3 is designed to deliver 99.999999999% (11 nines) of data durability. Objects are redundantly stored across multiple Availability Zones within an AWS Region. This durability applies to Standard, Standard-IA, Intelligent-Tiering, and other storage classes. Amazon S3 also provides 99.99% availability for the S3 Standard storage class.

Note: Amazon S3 stores data as objects within resources called buckets. An object consists of a file and any metadata that describes that file. A key uniquely identifies each object within a bucket.

Key Benefits

Scalability: Amazon S3 automatically scales to handle any amount of data, from a few gigabytes to exabytes, without upfront provisioning.

Durability: 11 nines of durability ensures your data is protected through redundant storage across multiple facilities.

Security: Server-side encryption, bucket policies, ACLs, S3 Block Public Access, and AWS IAM integration provide multi-layered security.

Cost efficiency: Multiple storage classes allow you to optimize costs based on access frequency and retrieval requirements.

Query in place: Amazon S3 Select and Amazon Athena let you run SQL queries directly on data stored in S3.

Core Concepts

Buckets

A bucket is a container for objects stored in Amazon S3. Every object is contained in a bucket. Bucket names must be globally unique across all AWS accounts and all AWS Regions within a partition. Bucket names can consist only of lowercase letters, numbers, and hyphens. A bucket name must start and end with a letter or number, and must be between 3 and 63 characters long.

Each bucket is created in a specific AWS Region, and objects stored in a bucket are never transferred outside that Region unless you explicitly move them. This is important for compliance and data residency

requirements.

Objects and Keys

An object is the fundamental entity stored in Amazon S3. Objects consist of object data and metadata. The metadata is a set of name-value pairs that describe the object. Some metadata is set by default, such as the date last modified, while others can be set by you, such as content-type. The object key (or key name) uniquely identifies the object in a bucket. The key is the full path of the object, for example: photos/2024/january/vacation.jpg.

Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in an S3 bucket. With versioning, you can recover from both unintended user actions and application failures. When you enable versioning on a bucket, Amazon S3 automatically assigns a unique version ID to each object stored.

Objects that existed before you enable versioning have a version ID of null. Suspending versioning does not delete existing object versions; it only prevents new versions from being created for new PUT operations.

Storage Classes

Amazon S3 offers a range of storage classes designed for different use cases. Selecting the appropriate storage class can significantly reduce costs while maintaining the required availability and durability.

Storage Class	Availability	Min Duration	Retrieval Fee	Use Case
S3 Standard	99.99%	None	None	Frequently accessed data
S3 Intelligent-Tiering	99.9%	None	None*	Unknown/changing access
S3 Standard-IA	99.9%	30 days	Per GB	Infrequently accessed
S3 One Zone-IA	99.5%	30 days	Per GB	Recreatable infreq. data
S3 Glacier Instant	99.9%	90 days	Per GB	Archive, ms retrieval
S3 Glacier Flexible	99.99%	90 days	Per GB	Archive, min-hr retrieval
S3 Glacier Deep Archive	99.99%	180 days	Per GB	Long-term archive

*S3 Intelligent-Tiering has a monitoring and automation charge but no retrieval fee.

S3 Intelligent-Tiering

S3 Intelligent-Tiering is the only storage class that automatically moves data to the most cost-effective access tier based on access frequency, without performance impact or operational overhead. It works by monitoring access patterns and moving objects that have not been accessed for 30 consecutive days to the Infrequent Access tier. If the object is accessed later, it is automatically moved back to the Frequent Access tier.

S3 Intelligent-Tiering also offers optional asynchronous Archive tiers for objects that have not been accessed for 90 or 180 days, providing even greater cost savings for long-lived data.

Security and Access Control

Security is of utmost importance when storing data in Amazon S3. AWS provides multiple mechanisms to control who can access your S3 resources and what actions they can perform.

Block Public Access

S3 Block Public Access provides settings for buckets and accounts to help manage public access to Amazon S3 resources. By default, new buckets do not allow public access. Block Public Access settings override bucket policies and object ACLs, providing a simple way to ensure that public access is never granted. There are four independent settings:

- `BlockPublicAcls` – Blocks new public ACLs and uploading objects with public ACLs.
- `IgnorePublicAcls` – Ignores all public ACLs on the bucket and objects.
- `BlockPublicPolicy` – Blocks new bucket policies that grant public access.
- `RestrictPublicBuckets` – Restricts access to buckets with public policies to only AWS services and authorized users.

Bucket Policies and IAM

Bucket policies are resource-based policies written in JSON that you attach to a bucket to grant or deny access to the bucket and the objects in it. IAM policies can be attached to users, groups, or roles to grant or deny access to Amazon S3 actions and resources. When both an IAM policy and a bucket policy apply, AWS evaluates the union of the permissions and applies the least-privilege principle.

Server-Side Encryption

Amazon S3 supports multiple server-side encryption options to protect data at rest. Since January 2023, all new objects uploaded to S3 are automatically encrypted at no additional cost.

- `SSE-S3`: Amazon S3 manages encryption keys using AES-256 encryption.

- SSE-KMS: AWS Key Management Service (KMS) manages the encryption keys, providing additional control and audit capabilities.
- SSE-C: You manage the encryption keys; Amazon S3 performs the encryption and decryption.
- DSSE-KMS: Dual-layer server-side encryption with AWS KMS keys for compliance workloads requiring two independent layers of encryption.

Data Management Features

Replication

Amazon S3 replication enables automatic, asynchronous copying of objects across S3 buckets. Replication can be within the same AWS Region (Same-Region Replication, or SRR) or across different AWS Regions (Cross-Region Replication, or CRR). Replication requires versioning to be enabled on both the source and destination buckets.

Common use cases for replication include compliance and data residency requirements, minimizing latency for geographically dispersed users, disaster recovery, aggregating logs from multiple source buckets into a single destination, and maintaining a live replica in a different account for production and test environments.

S3 Object Lambda

S3 Object Lambda allows you to add your own code to S3 GET, HEAD, and LIST requests to modify and process data as it is returned to an application. You can use S3 Object Lambda to dynamically transform data for the requester, remove personally identifiable information (PII) for analytics, convert between data formats such as XML and JSON, resize and watermark images, and more.

S3 Batch Operations

S3 Batch Operations is a managed solution for performing storage actions such as Copy, Invoke AWS Lambda, Replace Object Tagging, Replace Access Control List, and Restore Initiation on large numbers of objects. You can use Batch Operations to make changes to billions of objects at once, without writing custom application code or managing the compute infrastructure to do so.

Performance Optimization

Amazon S3 is designed to support very high request rates. Your application can achieve at least 3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket. There is no limit to the number of prefixes in a bucket, so you can achieve virtually unlimited request rates.

Multipart Upload

Multipart upload allows you to upload a single object as a set of parts. Each part is a contiguous portion of the object's data. You can upload these object parts independently and in any order. If transmission of any part fails, you can re-transmit that part without affecting other parts. After all parts of your object are uploaded, Amazon S3 assembles these parts and creates the object.

Multipart upload is recommended for objects 100 MB or larger, and is required for objects exceeding 5 GB. The maximum object size in Amazon S3 is 5 TB.

Amazon S3 Transfer Acceleration

S3 Transfer Acceleration enables fast, easy, and secure transfers of files over long distances between your client and an S3 bucket. Transfer Acceleration takes advantage of Amazon CloudFront's globally distributed edge locations. As data arrives at an edge location, it is routed to Amazon S3 over an optimized network path.

Event Notifications and Integrations

Amazon S3 can be configured to send event notifications when specific events occur in your S3 bucket, such as object creation, deletion, or restoration. These notifications can be sent to Amazon Simple Notification Service (SNS), Amazon Simple Queue Service (SQS), AWS Lambda, or Amazon EventBridge.

With Amazon EventBridge, you can use advanced filtering, archiving, and replay capabilities to build sophisticated event-driven architectures. EventBridge supports all S3 event types and allows routing events to over 20 AWS services and third-party partners.

Lifecycle Policies

An S3 Lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. You can use lifecycle policies to transition objects between storage classes or expire objects at the end of their lifecycle. Lifecycle policies help you automate cost optimization without manual intervention.

Lifecycle rules can be applied to all objects in a bucket, or you can filter by object prefix, tag, or object size. Transition actions can move objects from Standard to Standard-IA after 30 days, then to Glacier after 90 days, and finally to Glacier Deep Archive after 180 days, for example.

Best Practices

Enable versioning for critical data: Versioning protects against accidental deletion and overwrites. Use MFA Delete for additional protection on versioning-enabled buckets.

Use S3 Intelligent-Tiering for unpredictable workloads: For objects with changing or unknown access patterns, Intelligent-Tiering automatically optimizes costs without performance impact.

Apply the principle of least privilege: Use IAM policies and bucket policies to grant only the permissions necessary for each user, role, or service.

Enable S3 Access Logs and CloudTrail: Audit access to your S3 resources by enabling server access logging and AWS CloudTrail data events for comprehensive audit trails.

Use VPC Endpoints for private access: Avoid sending S3 traffic over the public internet by using S3 Gateway Endpoints or S3 Interface Endpoints within your VPC.

Implement lifecycle policies: Define lifecycle rules to automatically transition data to lower-cost storage classes and expire objects that are no longer needed.

Use multipart upload for large objects: Always use multipart upload for objects over 100 MB to improve throughput and resilience to network interruptions.

Enable S3 Replication for disaster recovery: Cross-Region Replication provides geographic redundancy and supports your disaster recovery strategy.

Frequently Asked Questions

Q: What is the maximum object size I can store in Amazon S3?

A: The maximum object size is 5 TB. For objects larger than 5 GB, you must use multipart upload. A single PUT operation can upload objects up to 5 GB.

Q: How do I make objects publicly accessible?

A: To make objects publicly accessible, you must disable Block Public Access settings at the bucket or account level, then configure a bucket policy or object ACL that grants public read access. AWS recommends using CloudFront as a CDN instead of making S3 objects directly public.

Q: Can I run SQL queries on data stored in S3?

A: Yes. You can use Amazon S3 Select to retrieve a subset of data from an object using SQL expressions. For more complex analytics, use Amazon Athena, which can query data stored in S3 using standard SQL without loading it into a database.

Q: How does S3 pricing work?

A: S3 pricing is based on the amount of data stored (per GB per month), the number of requests made (PUT, GET, etc.), data transfer out of S3, and any optional features such as Replication, Intelligent-Tiering monitoring, or S3 Object Lambda. Storage costs vary by storage class.

Additional Resources

- [Amazon S3 Developer Guide](#)
- [Amazon S3 pricing page](#)
- [S3 Storage Classes overview](#)
- [Amazon S3 security best practices](#)
- [Amazon S3 performance guidelines](#)
- [AWS Well-Architected – Storage pillar](#)
- [Amazon S3 Replication documentation](#)